

AI Health Chatbot – Assignment Report

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AI-Based Health Q&A Chatbot with Multilingual Support

Introduction

In this Assignment,

I developed a simple AI Health Assistant Chatbot capable of:

- Answering user health queries
- Classifying queries into intent categories (Nutrition, Medicine, Fitness, Emergency)
- Supporting multilingual inputs for better accessibility

The chatbot aims to bridge the gap in health information availability by providing easy answers in both English and Hindi, ensuring wider reach in regional populations.

Tools & Languages Used

Programming Language: Python

Platform: GitHub , Visual Studio code

Documentation: Markdown

Languages: English, Hindi

Methodology

Step 1: Data Preparation

Created 10 health-related question-answer pairs covering topics like Nutrition, Medicine, Fitness, and Emergency queries.

Each Q&A pair was prepared in both English and Hindi to support multilingual users.

Like,

1. Q : What can I do to relieve a headache quickly?

A : Drink water, rest in a dark quiet room, and if needed, take Paracetamol.

Q : सिरदर्द जल्दी ठीक करने के लिए क्या करें?

A : पानी पिएं, अंधेरे शांत कमरे में आराम करें और जरूरत हो तो पैरासिटामोल लें।

2. Q : What foods help reduce high blood pressure?

A : Eat low-salt foods, green leafy vegetables, and bananas to reduce high BP.

Q : हाई ब्लड प्रेशर कम करने के लिए क्या खाएं?

A : कम नमक वाला खाना, हरी पत्तेदार सब्जियाँ और केला खाना फायदेमंद है।

3.Q ;What is a home remedy for cold and cough?

A : Drink ginger tea with honey and inhale steam with eucalyptus oil.

Q : सर्दी खांसी का घरेलू उपाय क्या है?

A : अदरक वाली चाय में शहद डालकर पिएं और यूकेलिप्टस ऑयल की भाप लें।

4.Q : How can I keep my skin healthy naturally?

A: Drink enough water, eat fresh fruits, and avoid junk food for healthy skin.

Q : त्वचा को नेचुरली हेल्दी कैसे रखें?

A : भरपूर पानी पिएं, ताजे फल खाएं और जंक फूड से बचें।

5.Q : How to reduce eye strain from screen time?

A: Follow the 20-20-20 rule: every 20 minutes, look at something 20 feet away for 20 seconds.

Q: स्क्रीन से आंखों में होने वाले दर्द को कैसे कम करें?

A : 20-20-20 नियम अपनाएं – हर 20 मिनट बाद 20 फीट दूर 20 सेकंड तक देखें।

6.Q : How can I reduce hair fall naturally?

A : Massage scalp with coconut oil, eat protein-rich foods, and avoid harsh shampoos.

Q : बाल झड़ने से कैसे रोकें?

A : नारियल तेल से सिर की मालिश करें, प्रोटीन युक्त खाना खाएं और हार्श शैम्पू का उपयोग न करें।

7.Q: What to do for stomach pain?

A: Drink warm water, eat light food, and rest. If pain persists, consult a doctor.

Q: पेट दर्द में क्या करें?

A : गुनगुना पानी पिएं, हल्का खाना खाएं और आराम करें। दर्द बने रहने पर डॉक्टर से मिलें।

8.Q: How can I improve my sleep quality?

A : Avoid screens before sleep, maintain a fixed sleep time, and keep the room dark.

Q: नींद अच्छी लाने के लिए क्या करें?

A: सोने से पहले मोबाइल/टीवी न देखें, रोजाना एक ही समय पर सोएं और कमरे को अंधेरा रखें।

9.Q: What to eat to recover from weakness after illness?

A: Eat light, protein-rich foods like dal, eggs, and drink fresh fruit juices.

Q : बीमारी के बाद कमजोरी दूर करने के लिए क्या खाएं?

A: दाल, अंडा जैसे प्रोटीन युक्त हल्का खाना और ताजे फलों का जूस पिएं।

10.Q: How can I gain weight safely?

A: Eat more nuts, dairy products, and do strength training exercises.

Q : सुरक्षित तरीके से वजन कैसे बढ़ाएं?

A : ज्यादा मेवे खाएं, दूध-दही लें और स्ट्रेंथ ट्रेनिंग करें

Step 2: Intent Classification

Developed a keyword-based Python function to classify user queries into predefined intents.

Code :

Simple Intent Classifier

```
def classify_intent(question):
```

```
    question = question.lower()
```

```
    # Define keywords per intent
```

```
    intent_keywords = {
```

```
        "nutrition": ["eat", "food", "diet", "banana", "hydration", "fruits"],
```

```
        "medicine": ["paracetamol", "antibiotic", "tablet", "medicine", "safe"],
```

```
        "emergency": ["fever", "dengue", "food poisoning", "emergency", "pain", "vomit"],
```

```
        "general_health": ["blood pressure", "baby", "signs", "symptoms"]
```

```
    }
```

```
    for intent, keywords in intent_keywords.items():
```

```
        for keyword in keywords:
```

```
            if keyword in question:
```

```
                return intent
```

```
    return "unknown"
```

Test Example

```
questions = [
```

```
    "What to eat when hemoglobin is low?",
```

```
    "Is paracetamol safe during pregnancy?",
```

```
    "What to do in case of food poisoning?",
```

```
    "Can diabetics eat bananas?",
```

```
    "What are early signs of dengue?",
```

```
]
```

```
for q in questions:
```

```
    print(f"Q: {q}")
```

```
    print(f"→ Intent: {classify_intent(q)}\n")
```

Results

The chatbot is able to:

Answer user health queries in both English and Hindi

Correctly classify queries into Nutrition, Medicine, Fitness, Emergency

Conclusion

Through this project, I learned:

Multilingual dataset preparation

Building simple intent classification logic using Python

This is a foundational step towards creating AI-based healthcare solutions that can be expanded using machine learning or NLP models in the future for better accuracy and real-time deployment.

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